

Assignment 3 - LLM Security

In this assignment, you will have the opportunity to apply the jailbreak techniques learned during the lecture to analyze and attempt to **jailbreak your own group project**. The goal is to understand potential risks and develop strategies to mitigate them, ensuring the security and robustness of your project.

There are three tasks you need to complete. Tasks 1 and 2 cover jailbreaking and defense (2% and 3% of the final grade, respectively). Task 3 is mandatory and covers evaluation design and evidence.

Task 1 (1%)

In this task, you will apply the jailbreak techniques covered in the lecture to your group's AI project. Your aim in this exercise is to attempt to design a textual or visual prompt that causes your group project model to return something totally different from what you intended to get.

You should upload:

- Your input prompt **AND** the response received from your system.
- A short paragraph describing what technique you used, **AND** why the response is irrelevant to your project aim (up to 200 words).

Alternatively, if you have already deployed strong defense measures in your project and thus cannot jailbreak your project for now, you can also upload (you will not get full marks for this and the following task if your attack is too weak when choosing this alternative):

- Your input prompt **AND** the response received from your system.
- A short paragraph describing what technique you used, **AND** what the **expected failing response** from your system is.

Task 2 (2%)

In this task, you will develop and implement strategies to defend against the jailbreak techniques identified in Task 1. The goal is to enhance the security of your project by modifying the code or prompt design to prevent unintended outputs.

You should upload:

- Updates you've made to code or prompt design.
- The response received from your system when prompted identically as in Task 1.
- Write a short description explaining why the implemented strategy is effective (up to 200 words).

If you chose the alternative in Task 1, you should instead upload:

- The snippet of the code/prompt in your system that you think is relevant for defending your attack in Task 1.
- Explain your claim (up to 200 words).

Task 3 (2%) – Evaluation Design and Evidence

You have built and defended your system in Tasks 1 and 2. Task 3 asks one question: how do you know it works? Your goal is to run a small, structured evaluation and report your findings as evidence, not as informal impressions.

Step 1 – Pick what you are testing and how

Choose one specific thing your system should be able to do reliably, then choose the evaluation method that fits. Both decisions go in the same section of your report.

System type	What to test	Recommended method
Recommender system	Are suggestions relevant to what the user asked for?	Test dataset (20–30 examples)
Chatbot / assistant	Does it stay on topic and refuse out-of-scope queries?	User-based (5+ people)
Content generation	Does the output match the format or facts it was given?	Test dataset (20–30 examples)
RAG pipeline	Does the answer reflect what was retrieved?	Test dataset (20–30 examples)

User-based:	At least 5 people interact with your system using the same prompts. Each person rates every response on a simple 1–3 scale that you define in advance.
Test dataset:	Build 20 to 30 input examples before running the system. Each example needs an input, a description of the correct behavior, and a PASS / FAIL judgment you apply afterward.

Step 2 – Write your success threshold before you start

Before collecting any data, write down the number you would consider a good result. This is the most important step. If you decide what counts as success after seeing the results, the evaluation proves nothing.

Example:	<i>“At least 75% of test cases should receive a PASS.”</i>
Example:	<i>“At least 4 out of 5 users should rate each response a 2 or higher on my scale.”</i>
Example:	<i>“The system should correctly refuse off-topic queries in at least 8 out of 10 cases.”</i>

Write your threshold in one sentence and include it at the top of your results section.

Step 3 – Run the evaluation and fill in the results table

Run your evaluation and report the numbers. Fill in the table below as part of your submission. Do not replace it with a paragraph of text.

What you tested	Your threshold (Step 2)	Your result
<i>e.g. Relevance of recommendations</i>	$\geq 75\%$ PASS	18 / 24 = 75% ✓
<i>e.g. Off-topic refusal</i>	$\geq 80\%$ correct decline	6 / 10 = 60% ✗

Step 4 – Document failures and answer two questions

Find two cases where your system did not meet the criterion, and one case that is unusual or adversarial. For each, fill in this template:

Field	What to write
Input	The exact prompt or query you gave the system.
System output	What the system actually produced.
Problem	Which criterion was violated, in one sentence.
Likely cause	Your best guess at why this happened. Was it the prompt wording, the model, or a missing defense from Task 2?

After the three cases, answer these two questions in two to three sentences each:

Q1.	If you tested 200 examples instead of 20, do you think your pass rate would go up, go down, or stay the same? Why?
Q2.	Name one type of user input your evaluation did not cover at all. Why does that gap matter?

What to Submit:

1. A PDF file containing your answers to both tasks.

For Task 3, additionally submit:

2. One sentence: what you tested. One sentence: which method you used and why.
3. Your success threshold, written before you ran the evaluation.
4. Results table with counts and percentages.
5. Three failure cases using the template above (2 failures + 1 unusual input).
6. Two questions answered in 2–3 sentences each.

Marking Criteria:

Task 1:

1. Effective jailbreaking Or Proper jailbreaking efforts.
2. Reasonable and technically correct description.

Task 2:

1. Effective defense Or A proper choice of a snippet that is relevant to the defense.
2. Reasonable and technically correct explanation.

Task 3:

1. Evaluation method is appropriate and threshold is stated before results are collected.
2. Results table is complete with counts, percentages, and pass/fail indicators.
3. Failure cases are documented with input, output, problem, and likely cause.
4. Both reflection questions are answered with specific reasoning.